Module 3

VM Migration and Load Balancing

The VM lifecycle consists of several key stages:

- ✔ **Creation**: Involves the instantiation of a virtual machine from a template or disk image. During creation, administrators define parameters such as CPU, memory, storage, and networking configurations.
- ✔ **Provisioning**: Refers to the allocation of resources to the newly created VM. This may involve attaching virtual disks, assigning IP addresses, configuring network settings, and installing necessary software components.
- ✔ **Virtual Machine migration (optional)** : Moving the VM's state to a different physical host for various reasons
- ✔ **Deployment**: Involves placing the VM into production or test environments. Deployment strategies may vary depending on the intended use case, such as development, testing, staging, or production environments.
- ✔ **Operation**: Encompasses the day-to-day management of VMs, including monitoring performance, managing resources, applying patches and updates, and troubleshooting issues.
- ✔ **Monitoring**: Involves tracking VM performance metrics such as CPU utilization, memory usage, disk I/O, and network traffic to identify potential bottlenecks or anomalies.
- ✔ **Scaling**: Refers to adjusting the capacity of VMs to accommodate changes in workload demand. Scaling may involve adding or removing resources dynamically or migrating VMs to different hosts.
- ✔ **Decommissioning**: Involves retiring VMs that are no longer needed or have reached the end of their lifecycle. Decommissioning may include archiving data, reclaiming resources, and removing VMs from the environment.

Process and System Level VMs:

- Process-level VMs: These virtualization techniques isolate individual processes or applications within a virtual environment, providing enhanced security, fault tolerance, and resource management. Examples include containers (e.g., Docker) and application-level virtualization technologies.
- System-level VMs: System-level virtualization isolates entire operating systems, allowing multiple operating systems to run concurrently on a single physical machine. Examples include hypervisors like VMware ESXi, Microsoft Hyper-V, and open-source solutions like KVM (Kernel-based Virtual Machine).

VM Migration

Needs for VM Migration

- Upgrading
- Load balancing through resource usage
- Avoid VM failures

VM Migration techniques

- Live/Hot migration – Migration of one VM from physical host to another when the powered is ON and after source host state copied to target host. Source host suspended and entire VM Moves to target host. Live migration requires shared storage or mechanisms for synchronizing memory and storage state between hosts.

- Regular/Cold migration- Migration of one VM from physical host to another when the powered is OFF. The configuration file, Log file and disk of VM are moved to the target host. Cold migration may result in longer downtime but can be performed without shared storage.

VM Scaling:

VM scaling ensures that applications have adequate resources to meet performance requirements while optimizing resource utilization and minimizing costs. There are two main types of VM scaling:

- Vertical Scaling (Scaling Up): Involves increasing the capacity of a single VM by adding more CPU cores, memory, or storage resources. Vertical scaling is suitable for applications with increasing resource demands.

- Horizontal Scaling (Scaling Out): Involves adding more VM instances to distribute the workload across multiple servers or nodes. Horizontal scaling improves fault tolerance, scalability, and performance by leveraging distributed computing resources.